https://doi.org/10.5664/jcsm.10962

JCSM | Journal of Clinical Sleep Medicine

COMMENTARY

The evolution of REM sleep without atonia diagnostic thresholds in isolated REM sleep behavior disorder

Commentary on Leclair-Visonneau L, Feemster JC, Bibi N, et al. Contemporary diagnostic visual and automated polysomnographic REM sleep without atonia thresholds in isolated REM sleep behavior disorder. *J Clin Sleep Med*. 2024;20(2):279–291. doi:10.5664/jcsm.10862

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In this issue of the *Journal of Clinical Sleep Medicine*, Leclair-Visonneau et al addressed the critical aspect of diagnosing isolated rapid eye movement (REM) sleep behavior disorder (iRBD) and discriminative characteristics of muscle activity.[1] REM sleep behavior disorder (RBD) is a near-perfect predictor of future alpha-synucleinopathy with a 96.6% probability of overt neurodegenerative disease phenoconversion at 14 years,[2] surpassing known alpha-synucleinopathy prodromal biomarkers. This study greatly expands our understanding of a powerful biomarker, REM sleep without atonia (RSWA), which provides hope in weathering the storm ahead as the elderly population (over 65 years) is expected to reach 80 million (22% of the population) by 2040.[3] As the U.S. population continues graying, a looming crisis in neurodegenerative disorders is expected, emphasizing the importance of identifying biomarkers for neuroprotective/disease-modifying treatment trials.[4] In this study, the authors broke new ground, provided insightful data, explored new variables, and confirmed/reinforced prior findings.

The scoring methodology for RSWA has undergone a multitude of modifications since its inception in 1992 (Montreal montage).[5–7] The RBD diagnostic thresholds of RSWA are based on the percentage of RSWA on video polysomnogram. According to the *International Classification of Sleep Disorders* (third edition) text revision using the American Academy of Sleep Medicine (AASM) scoring criteria, the optimal sensitivity and specificity cut-offs for any phasic or tonic RSWA in submentalis (SM) are more than 9% and 18%, respectively.[7] The RSWA thresholds for any tonic or phasic SM + flexor digitorum superficialis (FDS) were 27% and 36% to 45% for SM + tibialis anterior (TA).[8] RSWA metrics were scored using the Mayo Clinic and AASM criteria in this study.

This study (20 participants with iRBD vs 20 controls) measured primary outcomes using a 3-second mini-epoch scoring (Mayo Clinic method). Optimal thresholds/cut-off points were 6.5% for "any" SM and 15.1% for "any" SM+FDS, with the highest specificity/discriminatory cut-off metrics being 14% and 27.4% for the same muscles. The visually scored combined FDS "phasic" showed the best area under the curve at 96.4% (AASM criteria). For automated scoring with the REM atonia index, left FDS (96.7%), averaged bilateral FDS (92.8%), and SM (91%) showed excellent area under the curve values. The automated Ferri's REM atonia index[9] demonstrated better discriminatory values for SM and FDS but not for TA, which is consistent with visual scoring. In this context, it is important to highlight that the REM atonia index has been exclusively suggested for analyzing chin/SM elecgtromyography (tonic muscle) in prior studies. Its utilization for quantifying RSWA in "phasic" muscles, such as FDS and TA, as performed in this study, represents a noteworthy extension of the REM atonia index. This warrants further investigation and independent confirmation in future research. Of note, the tonic muscle activity of FDS and TA did not exhibit discriminative value in this study.

The study raises a crucial concern regarding the impact of varying RSWA quantification methodologies. The AASM methodology was shown to have the highest accuracy (92.2%) and specificity (99%) in detecting iRBD (92.2%)[10]; the SINBAR method has slightly higher sensitivity (93%) than AASM (92%)[10] likely due to 3-second mini epoch analysis. Researchers and clinicians must carefully consider the trade-off between employing diverse methodologies to understand novel REM atonia metrics and maintaining homogeneity for future meta-analytical purposes. There is a need for broader discussion within the scientific community regarding the standardization of methodologies while acknowledging the need for flexibility in adapting to novel advancements. Indeed, as technology continues to advance, incorporating automated scoring methods, such as the automated Ferri's REM atonia index, becomes increasingly relevant. Continued exploration of technological advancements in RSWA monitoring, and scoring could further refine diagnostic methodologies and contribute to the evolving landscape of artificial intelligence.

The inclusion of periodic limb movements during REM sleep in the scoring methodology is an intriguing aspect of this study, differing from the AASM and SINBAR approaches that exclude periodic limb movements.[6,7] A few studies have examined periodic limb movements in REM sleep, analyzing distinct characteristics that differentiate individuals with RBD from normal controls and patients with restless legs syndrome.[11–13] Nevertheless, additional clarification is warranted concerning the categorical inclusion or exclusion of periodic limb movements in the calculation of pathological thresholds for RSWA. Elaborating on this topic could provide insights into the potential relevance of TA movements in the context of iRBD diagnosis.

The study encourages exploration of optimal cut-offs for RBD diagnosis in specific subgroups, such as presymptomatic and secondary RBD patients. Furthermore, investigating cut-offs with maximized specificity to identify prodromal asymptomatic isolated RSWA (iRSWA) presents a promising direction for future research, offering an opportunity for early intervention in the course of alpha-synucleinopathy. These recommendations pave the way for a more nuanced understanding of RSWA across different clinical scenarios, contributing to the ongoing refinement of diagnostic criteria.

While acknowledging the notable strengths of the study, such as meticulous participant selection and comprehensive analysis of muscle activity characteristics, it is essential to recognize certain limitations. The relatively small sample size (20 participants with iRBD vs 20 controls) warrants caution in generalizing the findings to larger populations. The authors acknowledge the potential for type 2 error due to this limitation. Future research with larger and more diverse cohorts could further validate and extend the current findings.

The findings of this study have direct implications for clinical practice in the realm of sleep medicine. The established quantitative thresholds of RSWA for diagnosing iRBD provide clinicians with concrete and standardized measures, enhancing the precision and consistency of RBD diagnosis. Sleep medicine clinicians can now utilize specific muscle measurements, burst duration, and the REM atonia index as valuable tools in their diagnostic arsenal. In summary, Leclair-Visonneau et al's study significantly advances our understanding of diagnosing iRBD. While acknowledging its contributions, it also prompts critical reflections on the practical implications, methodological considerations, and avenues for future research.

ABBREVIATIONS

AASM, American Academy of Sleep Medicine
FDS, flexor digitorum superficialis
iRBD, isolated REM sleep behavior disorder
RBD, REM sleep behavior disorder
REM, rapid eye movement
RSWA, REM sleep without atonia
SM, submentalis
TA, tibialis anterior

CITATION

Gorantla S, Rodriguez C, Ferri R. The evolution of REM sleep without atonia diagnostic thresholds in isolated REM sleep behavior disorder. *J Clin Sleep Med*. 2024;20(2):187–188.

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Submitted for publication November 30, 2023
Accepted for publication November 30, 2023
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DISCLOSURE STATEMENT

All authors have reviewed the final version and approved the manuscript. The authors report no conflicts of interest.